

AI/ML Internship – Day 45 Notes & Tasks

Module 6: Installing YOLO & Running Your First Object Detection Project

Part 1: Recap of Day 44

Yesterday we learned:

-  Computer Vision
-  Object Detection
-  YOLO (You Only Look Once)
-  Bounding Boxes
-  Real-World Applications of YOLO

Today we will perform our first practical implementation using YOLOv8.

Part 2: What is YOLOv8?

Definition

YOLOv8 is one of the latest versions of YOLO developed by Ultralytics.

It is used for:

- Object Detection
 - Image Classification
 - Segmentation
 - Pose Estimation
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Why YOLOv8?

Compared to older versions:

-  Faster
-  More Accurate
-  Easier to Use
-  Better Performance

Part 3: Environment Setup

Step 1: Check Python Version

Open terminal and run:

```
python --version
```

Expected:

Python 3.10+

Step 2: Create Virtual Environment

```
python -m venv yolo_env
```

Activate environment:

Windows

```
yolo_env\Scripts\activate
```

Linux / Mac

```
source yolo_env/bin/activate
```

Part 4: Install YOLOv8

Install Ultralytics package:

```
pip install ultralytics
```

Verify Installation

```
yolo checks
```

Example Output

Python ✓

Torch ✓

CUDA ✓

Ultralytics ✓

Part 5: Understanding Pretrained Models

YOLO provides pretrained models.

Examples:

Model	Size
-------	------

yolov8n.pt	Nano
------------	------

yolov8s.pt	Small
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yolov8m.pt	Medium
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yolov8l.pt	Large
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yolov8x.pt	Extra Large
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Recommended for Beginners

yolov8n.pt

Reason:

- Fast
 - Small size
 - Easy to run
-

Part 6: Import YOLO

Create a file:

detect.py

Code:

```
from ultralytics import YOLO
```

Part 7: Load YOLO Model

```
from ultralytics import YOLO
```

```
model = YOLO("yolov8n.pt")
```

Explanation

YOLO → Algorithm

yolov8n.pt → Pretrained weights

The model already knows how to detect:

- Person
- Dog
- Cat
- Car
- Bus
- Bicycle

and many other objects.

🚀 Part 8: First Object Detection

Suppose we have:

image.jpg

Code:

```
from ultralytics import YOLO
```

```
model = YOLO("yolov8n.pt")
```

```
results = model("image.jpg")
```

```
results[0].show()
```

What Happens?

Input Image



YOLO Model



Object Detection



Bounding Boxes



Display Result

📌 Part 9: Detect and Save Output

```
from ultralytics import YOLO
```

```
model = YOLO("yolov8n.pt")
```

```
results = model("image.jpg", save=True)
```

Output

YOLO automatically creates:

```
runs/
```

```
└─ detect/
```

```
    └─ predict/
```

```
        └─ image.jpg
```

with detected objects.

📌 Part 10: Detect Objects from Multiple Images

```
from ultralytics import YOLO
```

```
model = YOLO("yolov8n.pt")
```

```
results = model([
```

```
    "image1.jpg",
```

```
    "image2.jpg",
```

```
    "image3.jpg"
```

```
], save=True)
```

📌 Part 11: Understanding Detection Output

Example Output:

Person 95%

Dog 92%

Car 88%

Confidence Score

Confidence indicates certainty.

Example:

Dog = 95%

Meaning:

Model is 95% sure it is a dog.

Part 12: Complete Program

```
from ultralytics import YOLO
```

```
model = YOLO("yolov8n.pt")
```

```
results = model(  
    "image.jpg",  
    save=True  
)
```

```
results[0].show()
```

Part 13: Real-World Example

Suppose an image contains:

1 Person

2 Cars

1 Dog

YOLO detects:

Person → Box

Car → Box

Car → Box

Dog → Box

All in a single scan.

Part 14: Common Errors

Error 1

ModuleNotFoundError

Solution:

```
pip install ultralytics
```

Error 2

Image not found

Solution:

Check image path.

Error 3

Python not recognized

Solution:

Install Python properly.

Part 15: Advantages of Pretrained Models

☒ No training required

☒ Ready to use

☒ Saves time

- ✓ Good accuracy
 - ✓ Suitable for beginners
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Part 16: Interview Questions

What is YOLOv8?

Latest YOLO object detection model.

What is yolov8n.pt?

Pretrained Nano model.

What is a pretrained model?

A model already trained on a large dataset.

What is a confidence score?

Confidence level of prediction.

Why use pretrained models?

To avoid training from scratch.

Tasks for Day 45

Theory Questions

1. What is YOLOv8?
 2. What is a pretrained model?
 3. What is a confidence score?
 4. Why is YOLOv8 popular?
 5. Difference between YOLOv8n and YOLOv8x?
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Practical Task 1

Install:

```
pip install ultralytics
```

Take screenshot.

Practical Task 2

Run:

```
yolo checks
```

Take screenshot.

Practical Task 3

Create:

```
from ultralytics import YOLO
```

```
model = YOLO("yolov8n.pt")
```

Execute successfully.

Practical Task 4

Download any image containing:

- Person
- Car
- Dog

Run object detection.

Practical Task 5

Record detected objects and confidence values.

Example:

Person → 97%

Dog → 94%

Car → 90%